

Matt's Timer

Native Android app. Countdown timer + stopwatch, built for back-to-back sets, and it keeps running with the app closed. Installs from dist/MattsTimer.apk - under 700 KB, no dependencies.

It cannot use the internet

There is no INTERNET permission in the manifest, so the phone will not let it reach the network under any circumstances. Nothing loads, nothing syncs, nothing needs signal. The chimes are synthesised on the device at startup rather than shipped as audio files.

The five permissions it does declare are all local to the phone: VIBRATE, and - so a set keeps running with the app closed - FOREGROUND_SERVICE, FOREGROUND_SERVICE_SPECIAL_USE, POST_NOTIFICATIONS and WAKE_LOCK.

The one rule that drives the layout

RESTART puts the clock back to the preset time and runs it immediately. Between back-to-back core sets that is one tap, not two. **RESET**, in the middle, does the same thing but leaves it stopped - it never starts anything.

Presets

35s 45s 1m 5m 10m 15m 20m 30m 60m - nine tiles, three by three.

The app **always launches on 35s**, regardless of what ran last - unless a set is still running, paused or finished, in which case it opens on that set. Presets go dim and inert while the timer is running, so a stray tap mid-set cannot wipe a live timer - pause or let it finish to change duration.

Tap the clock for a one-off time

With the timer stopped - ready, paused or finished - **tap the digits** and two wheels come up, minutes and seconds. Flick them, or tap a number and type it. SET & START applies the value and starts it in one tap, the same path RESTART takes; RESET and RESTART then come back to that value until a preset is tapped. CANCEL or the back button closes it, and the ready line reads TAP TO SET as the reminder. Tapping the clock while it is running, or in the stopwatch, does nothing.

The three buttons

Button	Timer	Stopwatch
Left	START / PAUSE / RESUME / GO AGAIN	START / STOP / RESUME
Middle - RESET	Back to the preset time, stopped	Back to zero, clears laps
Right	RESTART - back to the preset time and runs	LAP

Sound

Voice	Character
BELL	C major arpeggio landing on a full triad. Warm, the default
CHIME	Falling G-E-C resolving to an open fifth. Calmer, longer ring
PULSE	Three clipped tones then a high hold. Cuts through a loud gym

MUTE

Silent - vibration still fires

The header button cycles these and always shows the current one; each pick plays a short preview. Changing voice or muting mid-set takes effect on the next cue, with no restart.

At zero

Soft tick at 3, 2, 1 in the selected voice, then the finish cue rung twice. Screen goes red and flashes; the phone vibrates. Every tone is a sine with a soft attack and a natural decay, synthesised on the device - no buzzer, no audio files. Cues play on the alarm stream so they carry over gym noise, and vibration toggles separately. Both settings persist.

The music ducks under the last three seconds

Three seconds out the app asks the system for transient audio focus, so whatever is playing in the headphones drops under the 3-2-1 and the chime instead of fighting them, and **stays** down at TIME. It comes back up on its own the moment the set is reset or restarted - pausing hands it back too. The app never writes a volume; it only asks for the duck, which is what makes the return gradual and leaves every other app's volume alone.

It keeps running with the app closed

Starting a countdown - or the stopwatch - starts a foreground service, so the set survives leaving the app, the screen going off, and swiping the app out of recents. It sits on the lock screen as an ongoing notification that counts itself down, with the buttons for whatever state it is in: PAUSE and RESTART while running, RESUME and RESTART while paused, RESTART at TIME, STOP for the stopwatch. Tapping the notification opens the app on the live set, and the service stops itself the moment nothing is live.

A partial wake lock is held while the countdown runs, and for six seconds past the finish, so the chime and the buzz land on time with the screen off. Android 13 and up asks once, on the first start, whether the app may post notifications - the timer runs either way. Every change is written to the phone, so a killed process picks the set back up; a reboot clears it and the app opens fresh on 35s.

Stopwatch

Second tab in the header. START / STOP, LAP while running, RESET when stopped. Laps list split and cumulative time, newest first. Hundredths resolution. The screen stays awake while anything is counting and releases as soon as it stops; rotating the phone rebuilds the layout without disturbing a live set, or an open wheel picker.

Install on the phone

Easiest: plug the phone into the desktop over USB with USB debugging on, and the install runs from here in one command - no files to move by hand.

By hand: copy `dist/MattsTimer.apk` to the phone over USB, tap it in Files, and allow "install unknown apps" for the file manager the first time. The build is signed with the local debug key, which is fine for sideloading to your own phone.

Rebuilding it

Needs JDK 17 and the Android SDK (compileSdk 35). No AndroidX, no Material library, no third-party dependencies - framework views only, which is why the APK is under 700 KB.

```
cd android && ./gradlew assembleRelease
```

Bumping `versionCode` in `android/app/build.gradle.kts` is only needed if an install ever refuses to replace the previous one.